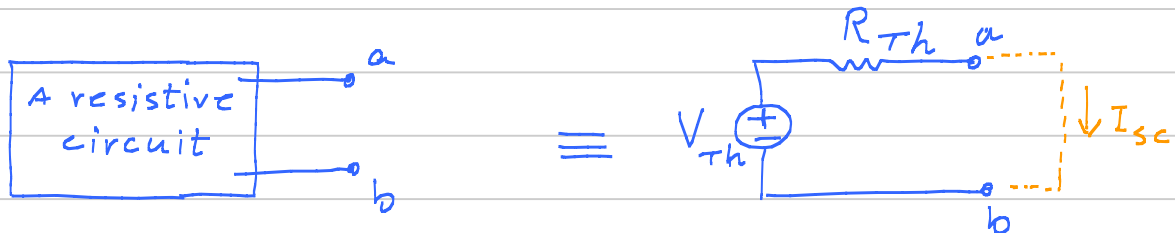




Thevenin Equivalent Circuit

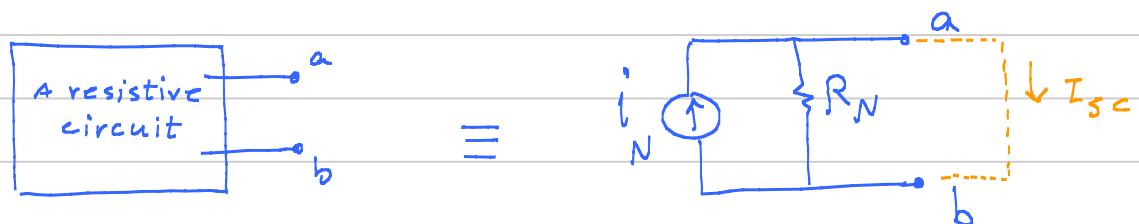
Thevenin equivalent circuit is a voltage source in series with a resistor which replaces a resistive network.



$$V_{Th} = V_{oc} \quad , \quad R_{Th} = \frac{V_{oc}}{i_{sc}}$$

Norton Equivalent Circuit

Norton equivalent circuit is a current source in parallel with a resistor which replaces a resistive network.



$$i_N = i_{sc} \quad , \quad R_N = \frac{V_{oc}}{i_{sc}}$$

To find Thevenin equivalent circuit:

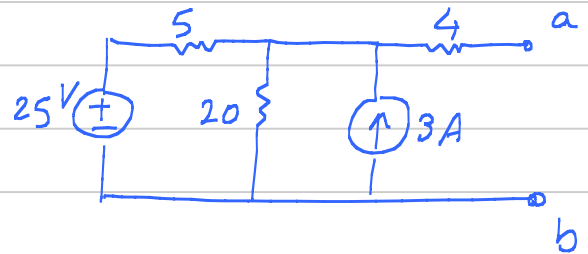
1. Find the **Thevenin voltage** by calculating the open circuit voltage.
2. Find the **Thevenin resistance** by calculating the ratio of the open circuit voltage which is the Thevenin voltage found in part 1, to the short circuit current.

If the circuit contains no dependent sources, the easiest way is to turn off all independent sources (To turn off an independent source, replace the independent current source with an open circuit and replace the independent voltage source with a short circuit). Then combine the resistors. The Thevenin resistance equals the equivalent resistance.

Ex. Find the Thevenin equivalent circuit.

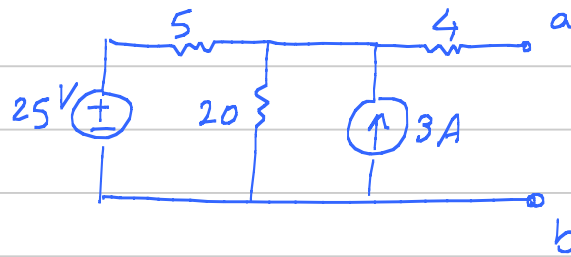
Finding V_{Th} :

$$V_{Th} = V_{oc}$$

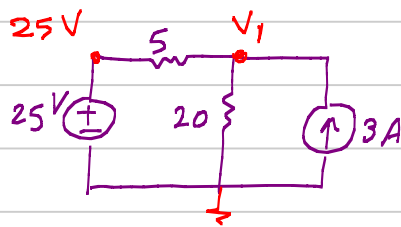


The open ckt. voltage is the voltage across the current source. we can use any method to find this voltage.

Let's use nodal analysis.



$$V_{Th} = V_{oc} = V_1$$

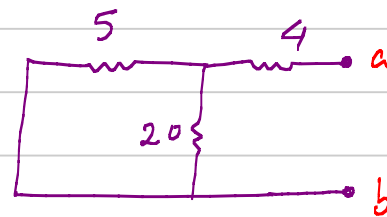


$$\frac{V_1 - 25}{5} + \frac{V_1}{20} - 3 = 0$$

$$V_1 = \boxed{32V} = V_{Th}$$

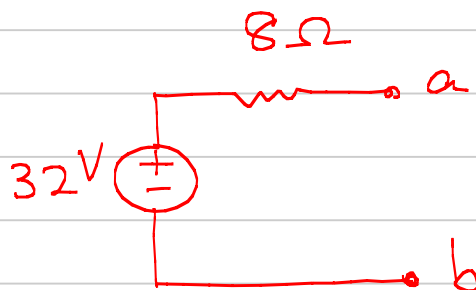
$R_{Th} = R_{eq}$ (when all independent sources are off)

Turn off indep. sources:

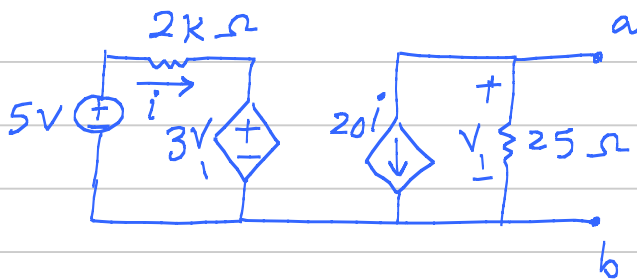


$$R_{eq} = 5 // 20 + 4 = \underline{8\Omega}$$

Therefore, the Thevenin equivalent circuit is:



Ex. Find the Thevenin equivalent circuit.



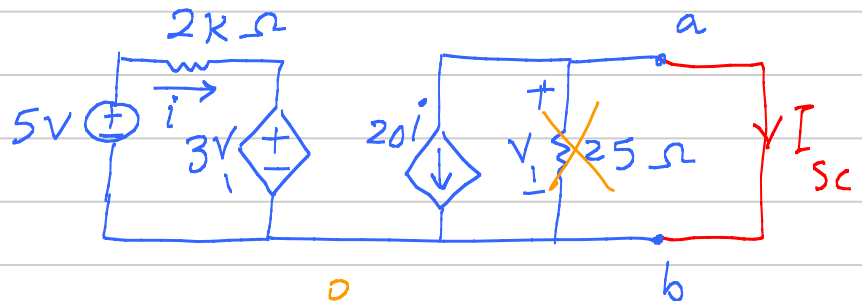
V_{Th}

$$V_{Th} = V_{oc} = V_1$$

$$V_1 = (25)(-20i) = -500i$$

$$V_1 = -500 \left(\frac{5 - 3V_1}{2000} \right) = \underline{-5V}$$

$$R_{Th} = \frac{V_{oc}}{I_{sc}}$$



$$V_1 = 0, \quad I_{sc} = -20i = -20 \left(\frac{5 - 3V_1}{2000} \right) = \underline{-\frac{5}{100} A}$$

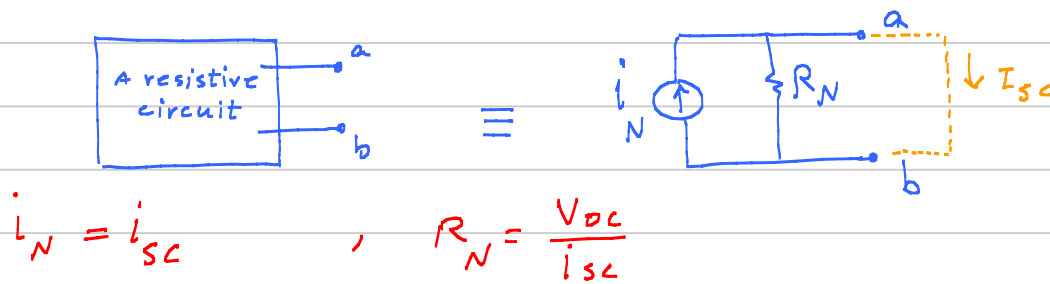
$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{-5}{-5/100} \Rightarrow \underline{100 \Omega}$$

Thevenin equivalent circuit:



Norton Equivalent Circuit

Norton equivalent circuit is a current source in parallel with a resistor which replaces a resistive network.



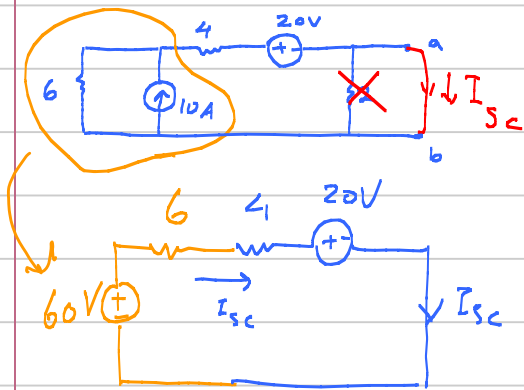
To find the Norton equivalent circuit:

1. Find the **Norton current**, I_N by calculating the short circuit current.
2. Find the **Norton resistance** by calculating the ratio of the open circuit voltage which is the Thevenin voltage found in part 1, to the short circuit current.

If the circuit contains no dependent sources, the easiest way is to turn off all independent sources (To turn off an independent source, replace the independent current source with an open circuit and replace the independent voltage source with a short circuit). Then combine the resistors. The Norton resistance equals the equivalent resistance.

Ex. Find the Norton equivalent circuit.

$$\underline{I_N = I_{sc}}$$

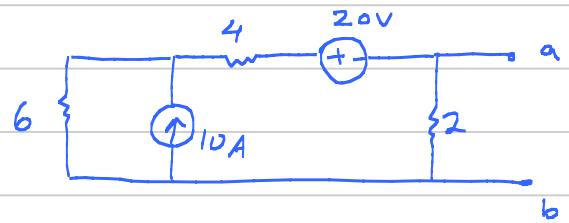
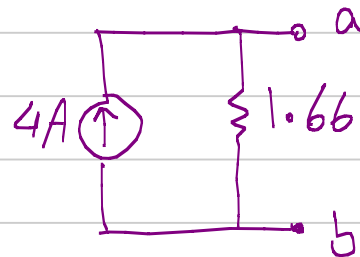


KVL:

$$-60 + 10 I_{sc} + 20 = 0$$

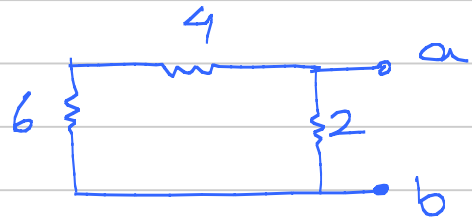
$$\underline{I_N = I_{sc} = 4A}$$

Norton equivalent circuit:



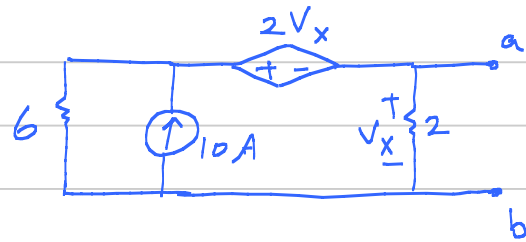
$$\underline{R_N}$$

Turn off all independent sources
and find the equivalent resistance.

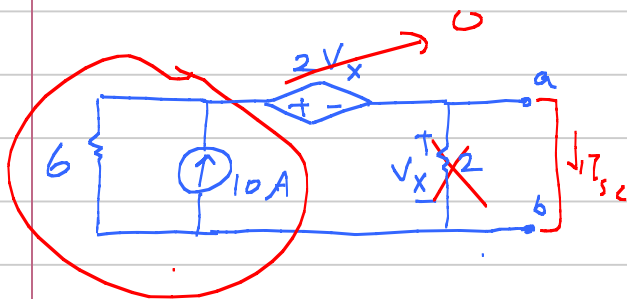


$$R_N = R_{ab} = (4+6) // 2 = \underline{1.66 \Omega}$$

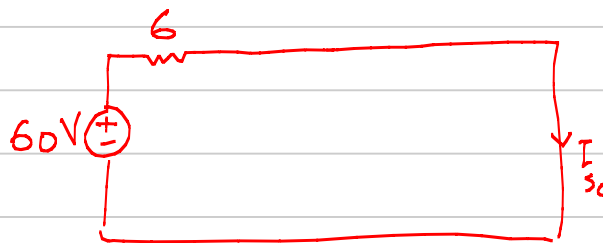
Ex. Find the Norton equivalent circuit.



I_N : $I_N = I_{sc}$

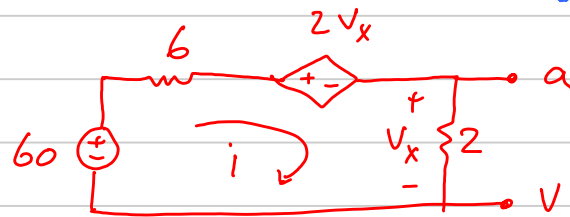
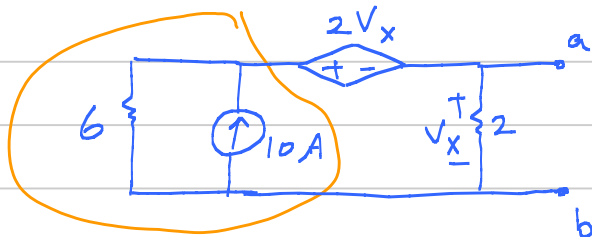


$$V_x = 0 \rightarrow 2V_x = 0$$



$$6 I_{sc} = 60 \rightarrow I_{sc} = \underline{10A}$$

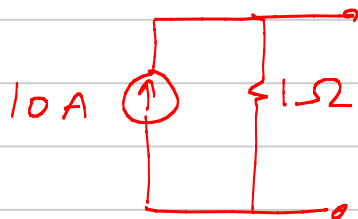
$V_{oc} = V_x$



$$\begin{cases} 6i + 2V_x + V_x - 60 = 0 \\ V_x = 2i \end{cases}$$

$$\Rightarrow V_x = 10V \Rightarrow V_{oc} = 10V$$

$$R_N = \frac{V_{oc}}{I_{sc}} = \frac{10}{10} = \underline{1\Omega}$$



⇐ Norton equivalent circuit